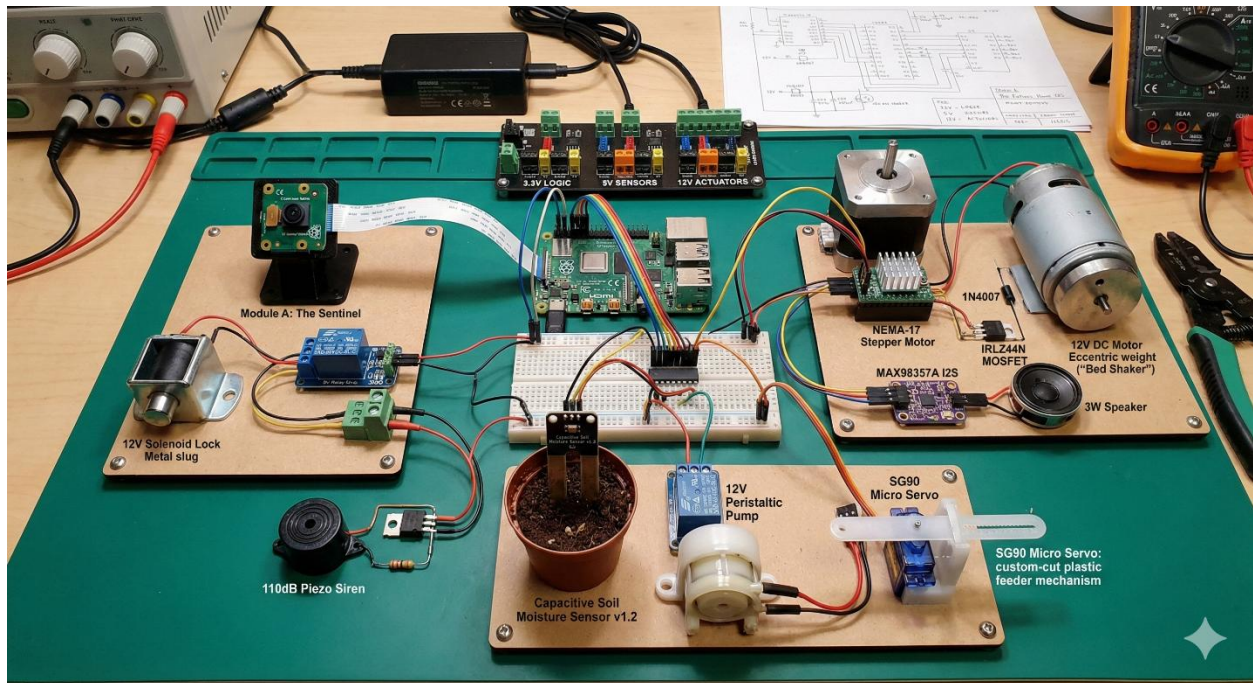




Hardware Architecture Visualization (AI-Generated Prototype)

This concept render illustrates the target wiring topology for "The Fortress" Home OS. It visualizes the physical interface between the central logic unit and the three active subsystems.

Visible Component Stack:

- **Central Hub:** Raspberry Pi 5 (GPIO & I2C Bus Master).
- **Module A (Sentinel):** 12V Solenoid Lock (Fail-Secure) driven by a 5V Relay and Pi Camera V2 for Computer Vision.
- **Module B (Caretaker):** NEMA-17 Stepper (Curtains) and High-Torque DC Motor with eccentric weight (Bed Shaker), driven via MOSFETs.
- **Module C (Sustainer):** Capacitive Soil Sensor (Analog) interfaced via ADC and a 12V Peristaltic Pump for precision irrigation.

Note: This image was generated using AI tools to visualize the final build schematic and component layout.